Host Utilities

TATA POWER

TATA POWER-DDL
Towards a Greener Tomorrow

BSES
BSES Rajdhani Power Limited

BSES
BSES Yamuna Power Limited

NPCL
NOIDA POWER COMPANY LIMITED

Knowledge Partner

Power Foundation of India

SESSION PARTNER

Digital Energy Solutions
Power Transmission & Distribution

ORGANIZER

ISGF
India Smart Grid Forum

Supporting Ministries

MINISTRY OF POWER
GOVERNMENT OF INDIA

NITI Aayog

MINISTRY OF POWER
GOVERNMENT OF INDIA
CENTRAL ELECTRICITY AUTHORITY

DEPARTMENT OF
SCIENCE & TECHNOLOGY

AIDA
Collaboration | Innovation | Excellence

India
SMART UTILITY
Week 2026
10 – 14 March 2026, New Delhi

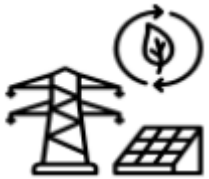
DIGITALIZATION ROADMAPS FOR UTILITIES

Presented By

SK SENAPATI, HEAD- SALES & COMMERCIALS, L&T-DIGITAL ENERGY SOLUTIONS

INDIA'S DIGITAL POWER TRANSFORMATION

Grid Intelligence & Autonomous Control



Digital Infrastructure for a Smart Power Ecosystem

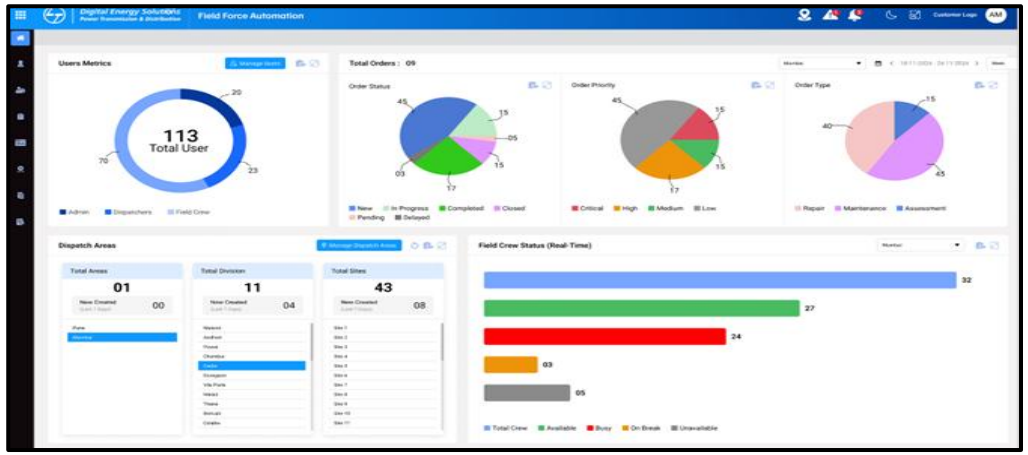
AI-Driven Optimization and Digital Energy Markets

Sensitivity: LNT Construction Internal Use

Grid Intelligence & Autonomous Control

ADMS & Field Force Automation (FFA)

Distribution, Outage management applications and Energy management with the **SCADA system** in a single UI.



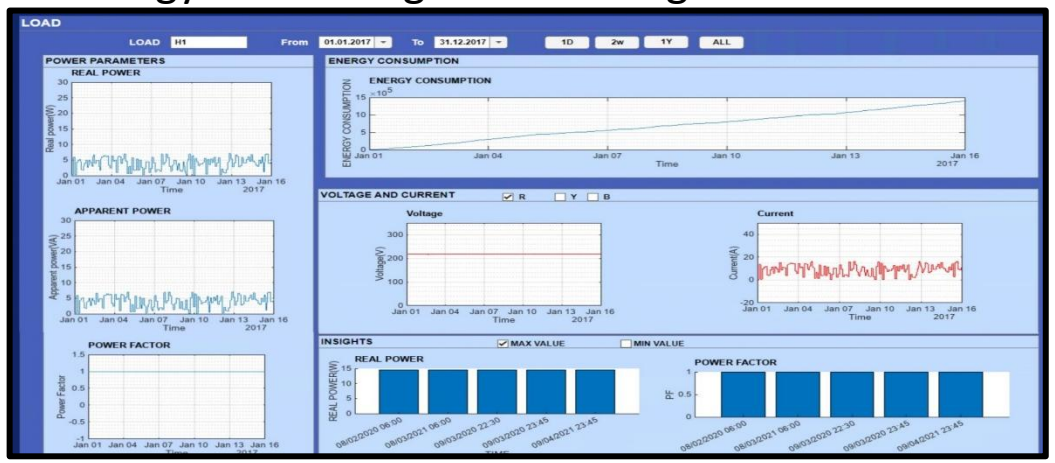
FFA - Field Force Automation

Advanced analytical tools and digital dashboards for field teams to improve responsiveness.

Theft Monitoring

Power Theft Detection

- Advanced analytics to detect tampering/anomaly
- Energy forecasting & scheduling tools



Smart Metering Solutions

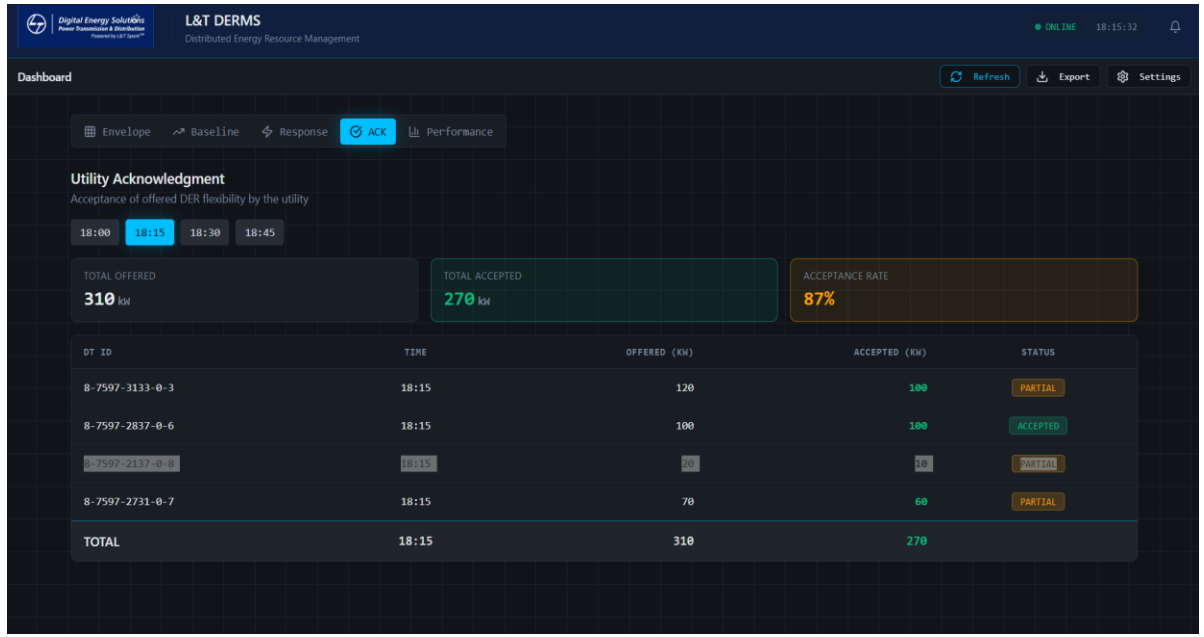
- Interoperability Framework
- Hybrid Connectivity Models
- Automated Data Validation

Sensitivity: LNT Construction Internal Use

Grid Intelligence & Autonomous Control

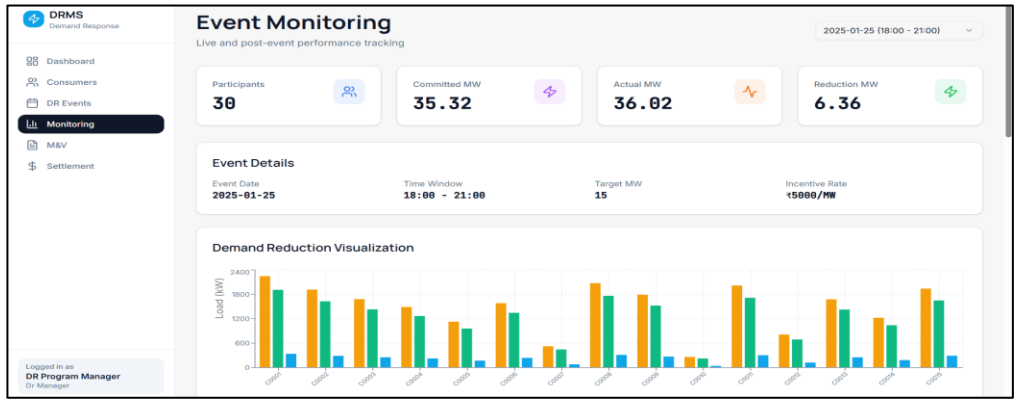
Distributed Energy Resource Integration Middleware (DERIM)

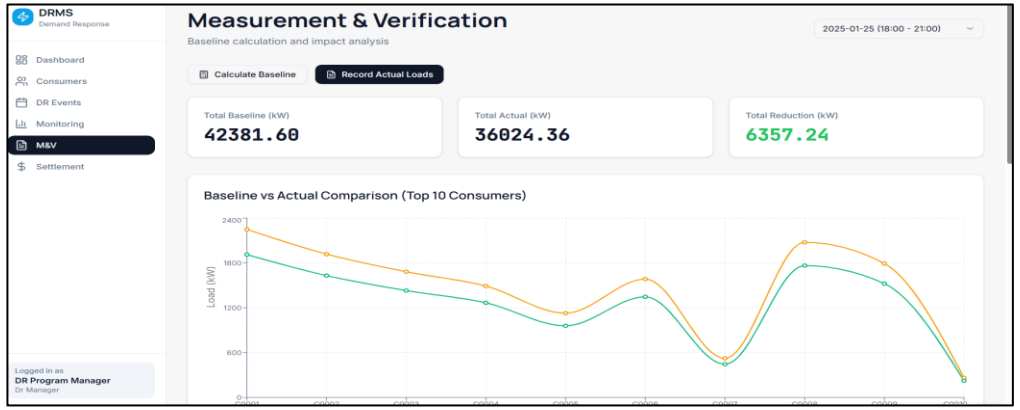
Distributed Energy Resource Integration Middleware (DERIM) acts as a **digital bridge** between DER assets, utility platforms, and market systems.



Demand Side Response (DSR)

Demand response provides **controllable flexibility** by allowing consumers to adjust consumption based on system needs.





Sensitivity: LNT Construction Internal Use

Blockchain-Enabled Peer-to-Peer(P2P) Energy Trading

The EMS enables **peer-to-peer trading** where prosumers sell surplus solar energy to local consumers through **bidding, matching, smart contracts, delivery, and meter-verified settlement**.

Technologies Enabling P2P Trading

- Smart Meters / AMI
- **Blockchain-based** energy marketplaces
- Real-time grid monitoring (SCADA)
- Digital platforms and mobile apps
- Distributed Energy Resource (DER)management

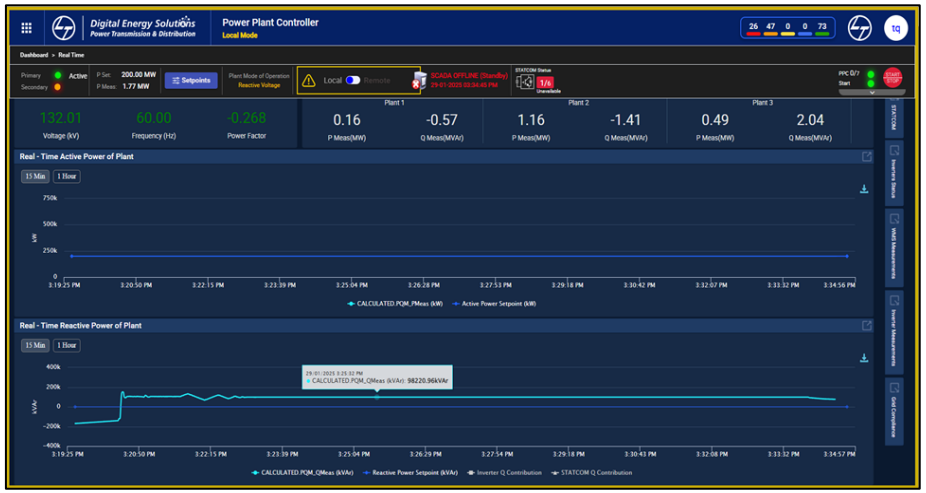
Sensitivity: LNT Construction Internal Use

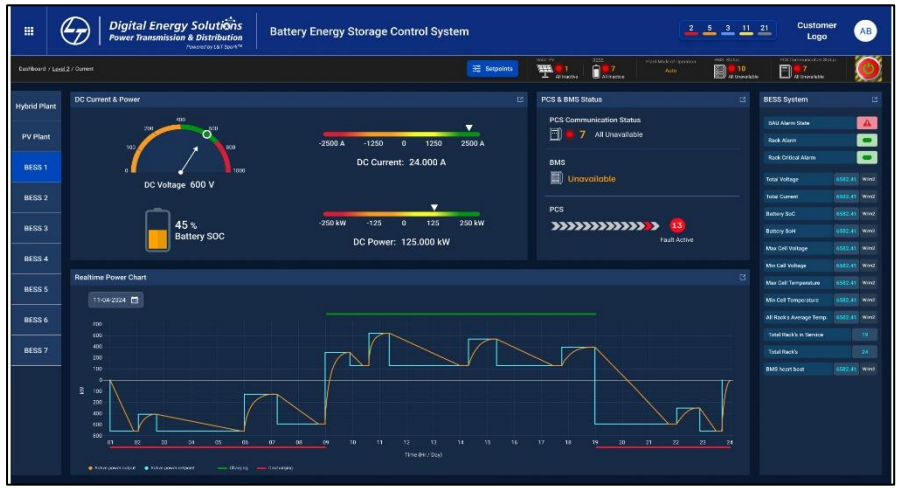
Digital Infrastructure for a Smart Power Ecosystem

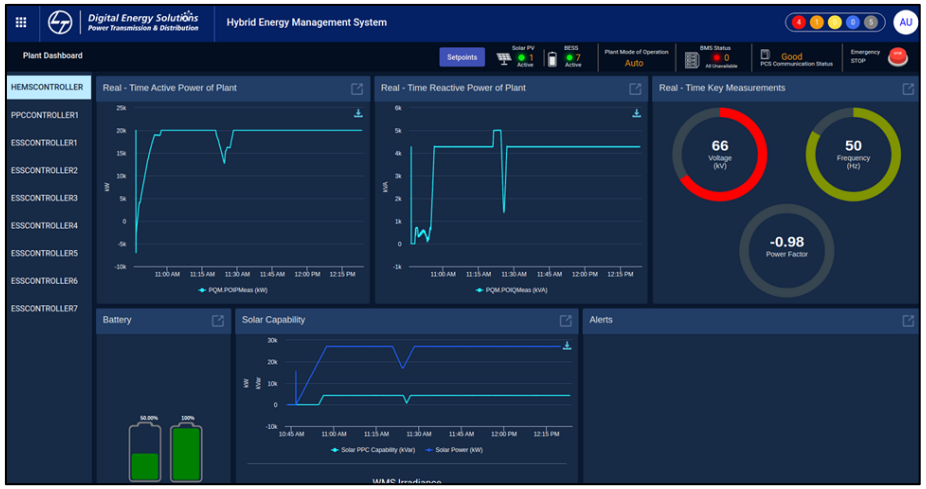
Renewable & Rooftop Solar Integration

Digital Stack:

- AI Driven Solutions
- Software Defined Solution
- Vendor Agnostics
- Advanced Analytics

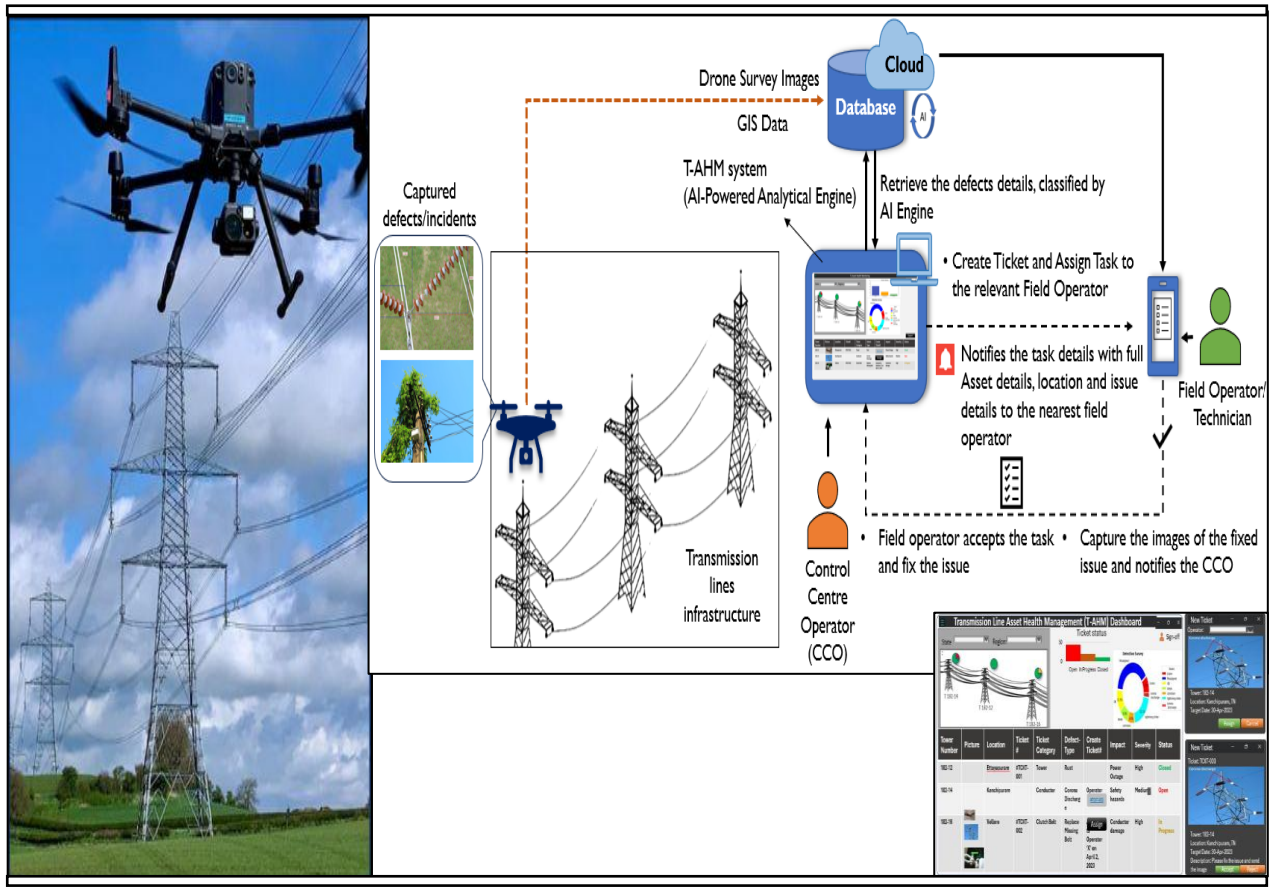






Sensitivity: LNT Construction Internal Use

Drone Based AI-Driven Transmission Line Health Management System



Integrating OT into the Security Operations Center (SOC)

Unified Visibility

Bridge Cultures

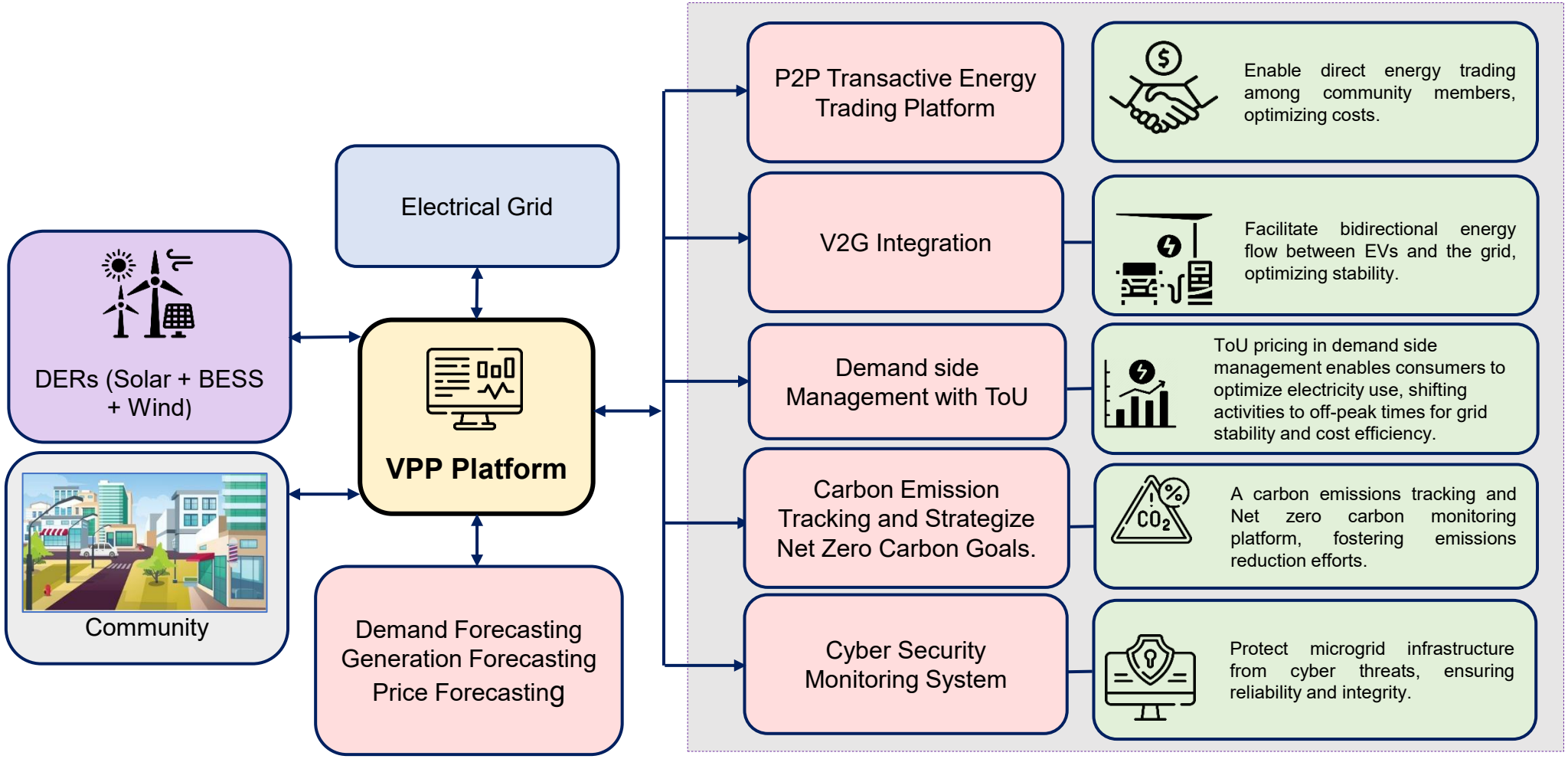
Phased Integration

Advanced SIEM platforms and threat intelligence systems provide comprehensive protection while maintaining operational integrity.

Sensitivity: LNT Construction Internal Use

Unified VPP platform

Foster a sustainable community with reduced emissions & overall energy expenses



Sensitivity: LNT Construction Internal Use

Future Roadmap

- AI-enabled grid analytics
- Digital twins for network planning
- Blockchain for energy trading
- Enhanced prosumer participation

Moving from Efficiency to Intelligence

1. Billing Efficiency
2. Operational Visibility
3. Predictive Intelligence
4. Autonomous Grid

Sensitivity: LNT Construction Internal Use

Collaborative Control Centers

Integrated Control Centers Enable:

- Unified situational awareness, simulation and control actions
- Shared dashboards across T&D systems
- Automated alerts & decision support

Sensitivity: LNT Construction Internal Use

Summary

Digitalization Roadmap Enables:

- Smart metering to analytics
- Grid intelligence via ADMS & DERIM
- Secure, integrated control centers
- AI-Driven Asset Health & Digital Workforce Tools
- Demand-side & market participation maturity

Sensitivity: LNT Construction Internal Use

Host Utilities

TATA POWER

TATA POWER-DDL
Towards a Greener Tomorrow

BSES
BSES Rajdhani Power Limited

BSES
BSES Yamuna Power Limited

NPCL
NOIDA POWER COMPANY LIMITED

Knowledge Partner

Power Foundation of India

SESSION PARTNER

Digital Energy Solutions
Power Transmission & Distribution

ORGANIZER

ISGF
India Smart Grid Forum

India SMART UTILITY Week 2026

10 – 14 March 2026, New Delhi

Supporting Ministries

MINISTRY OF POWER
GOVERNMENT OF INDIA

NITI Aayog

MINISTRY OF POWER
GOVERNMENT OF INDIA
CENTRAL ELECTRICITY AUTHORITY

DEPARTMENT OF
SCIENCE & TECHNOLOGY

AIDA
Collaboration | Innovation | Excellence

THANK YOU